

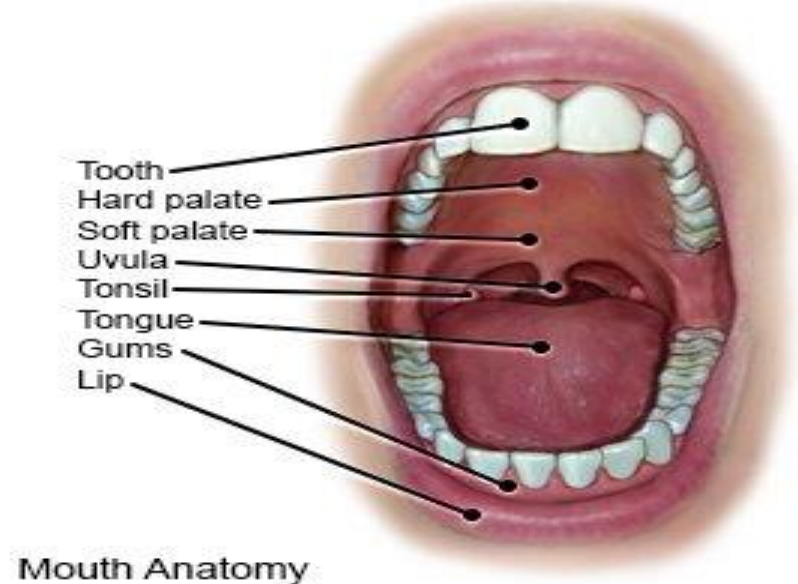
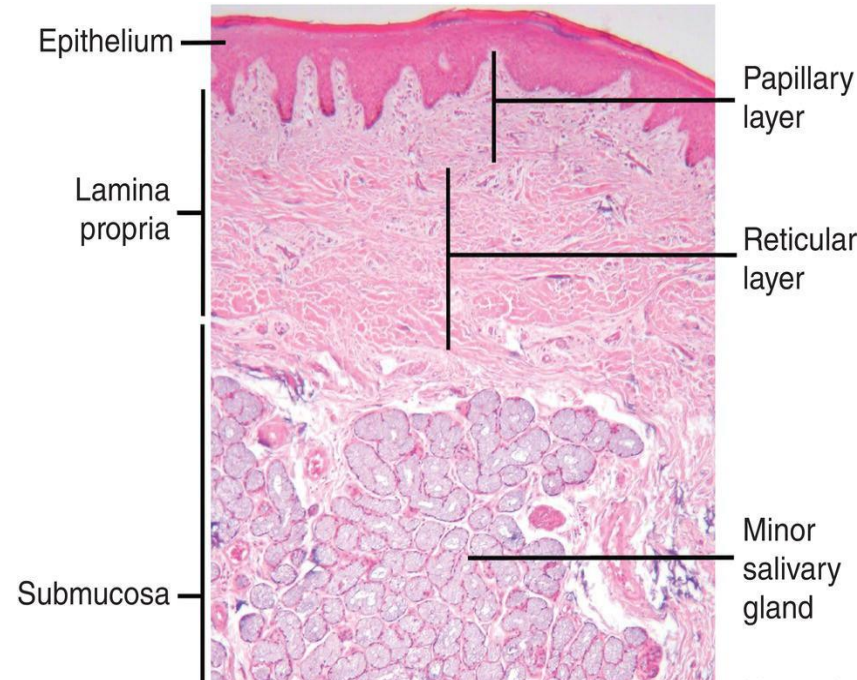
The Digestive System
oral cavity and salivary
glands
SGL1

Dr. Hero K. Mustafa

Oral Cavity (mouth): Initial site of mechanical digestion and chemical digestion by(saliva).

Is made up partly of bone (jaws, hard palate) & partly of muscle & connective tissue (lips, cheeks, soft palate & floor of mouth).

- lined by mucous membrane is stratified squamous epithelium which rests on connective tissue.
- In the cheeks, this connective tissue contains many elastic fibers & much fat.
- The roof of the mouth consists of the hard & soft palates, both covered by same type of epithelium.
- In the hard palate the mucous membrane rests on bony tissue. The soft palate has a core of skeletal muscle & numerous mucous glands in its submucosa.



Lips : Is skeletal muscle. • Each lip have an external surface & an internal surface :

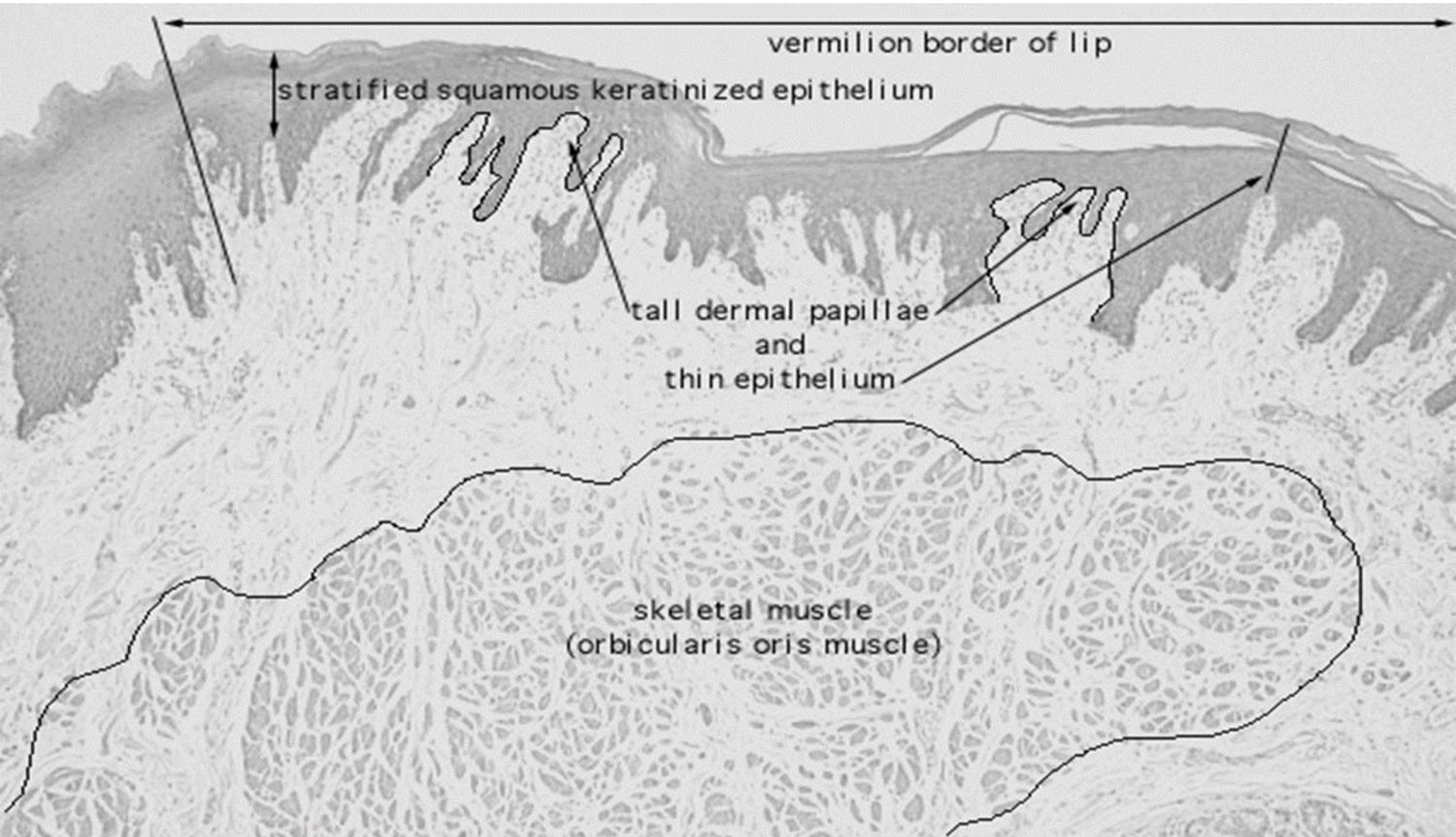
The **inner surface** of the lips lined by thick non keratinized st.sq.ep. & small clumps of salivary gland (labial gland), sebaceous gland.

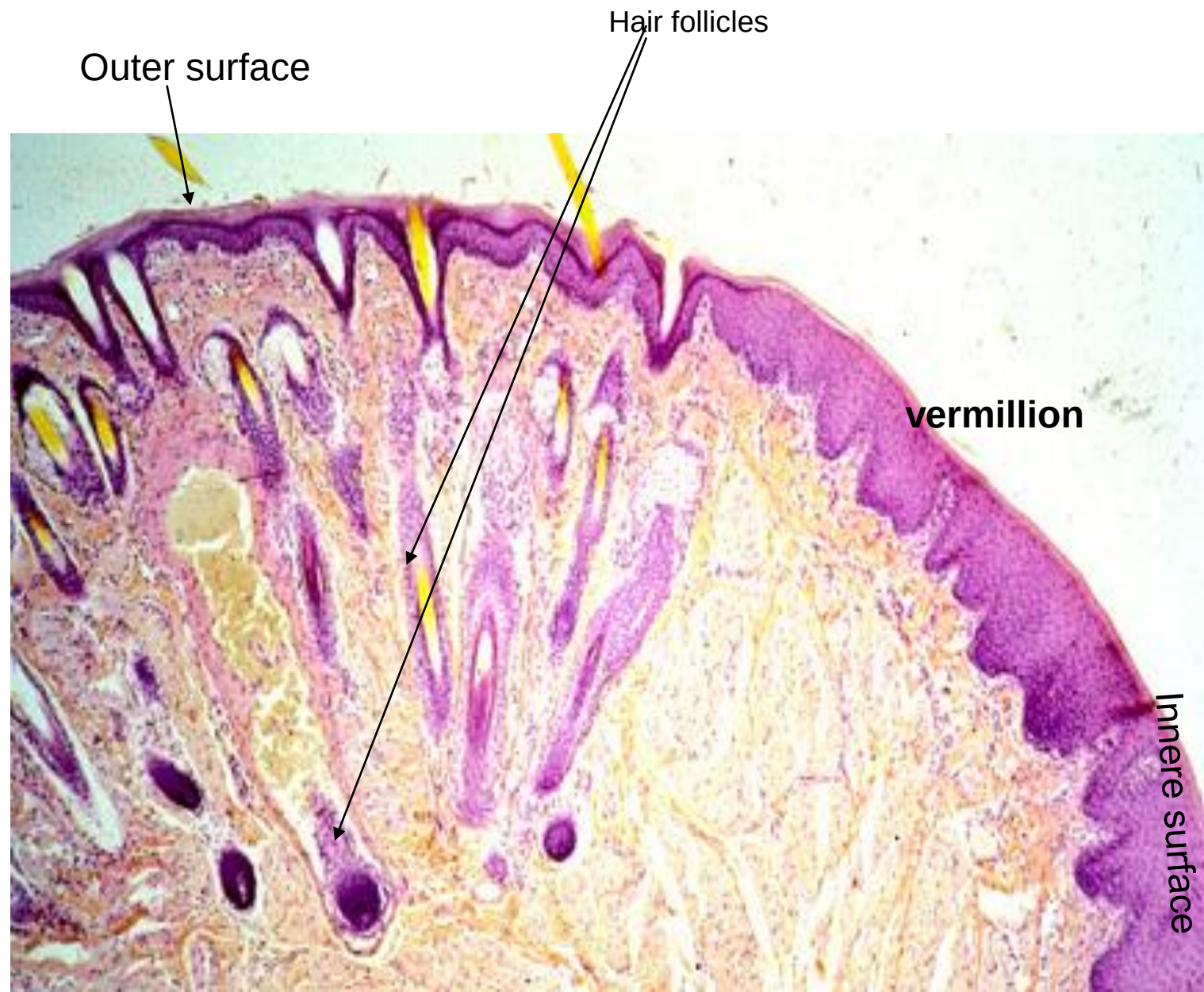
Outer surface of the lips lined by thin skin with hair follicles, sebaceous glands, and sweat glands.

Between the inner & outer surface of the lips there is a **transitional zone** known as **vermillion** border the epithelium is lightly keratinized s.s.ep., contain prominent blood vessels which gives it's red appearance with numerous sensory nerve endings , has no hair ,no sweat & sebaceous glands



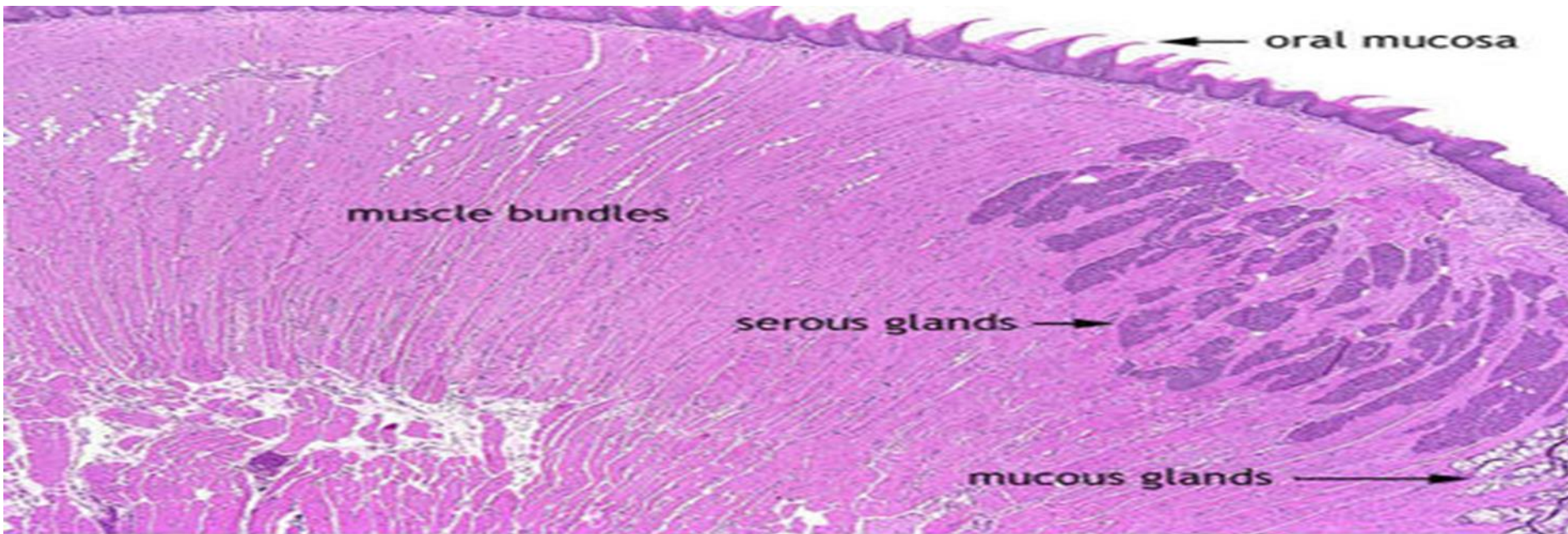
In the deeper parts of lips bundles of striated muscle fibers (**orbicularis oris muscle**) are arranged mainly in a concentric manner around the oral orifice





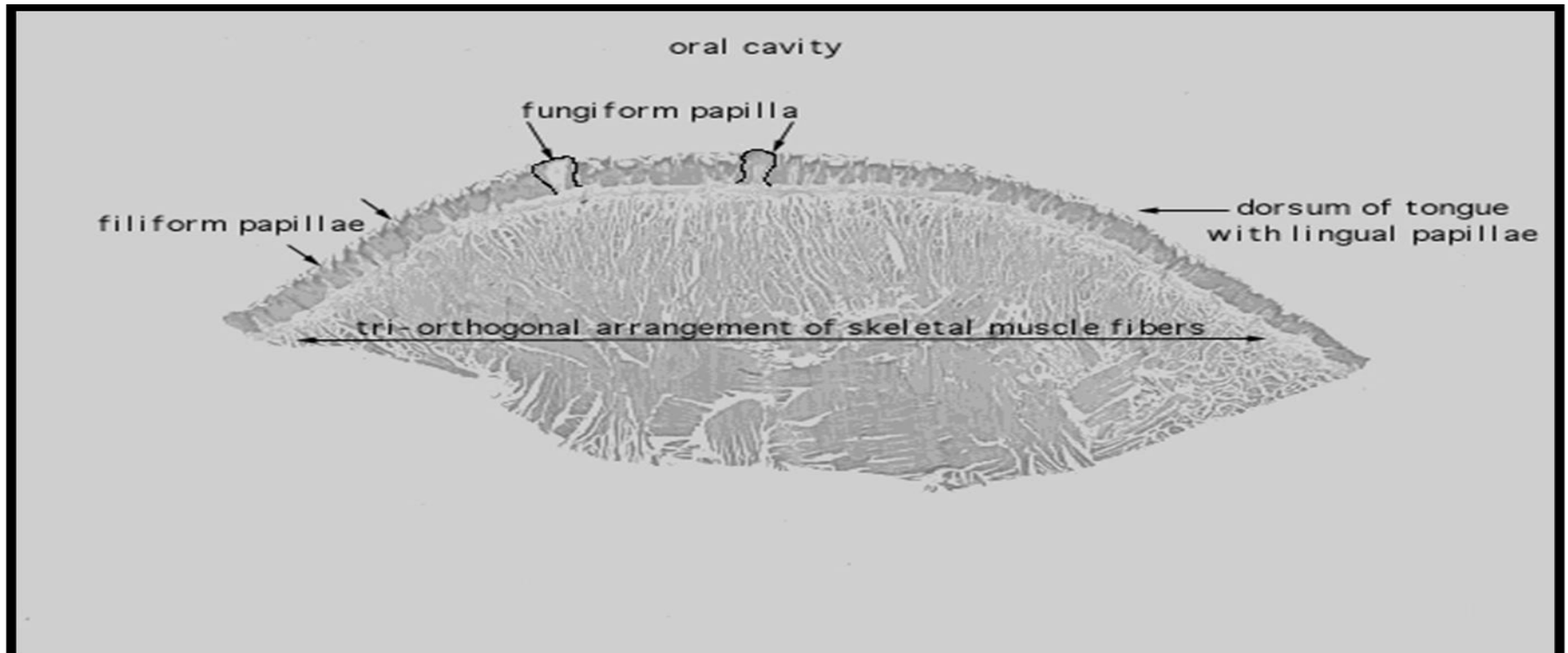
Tongue

- It's a mass of skeletal muscle occupies the floor of the mouth and fills the oral cavity.
- covered with lightly keratinized stratified squamous epithelium.
- The mucosa is bound tightly to the muscle by the lamina propria, the muscle is arranged in bundle of many size these are separated by connective tissue cross in 3 planes arranged in bundle this gives the tongue the flexibility for speech, positioning food, chewing & swallowing



The mucosa differ on the dorsal & ventral surface

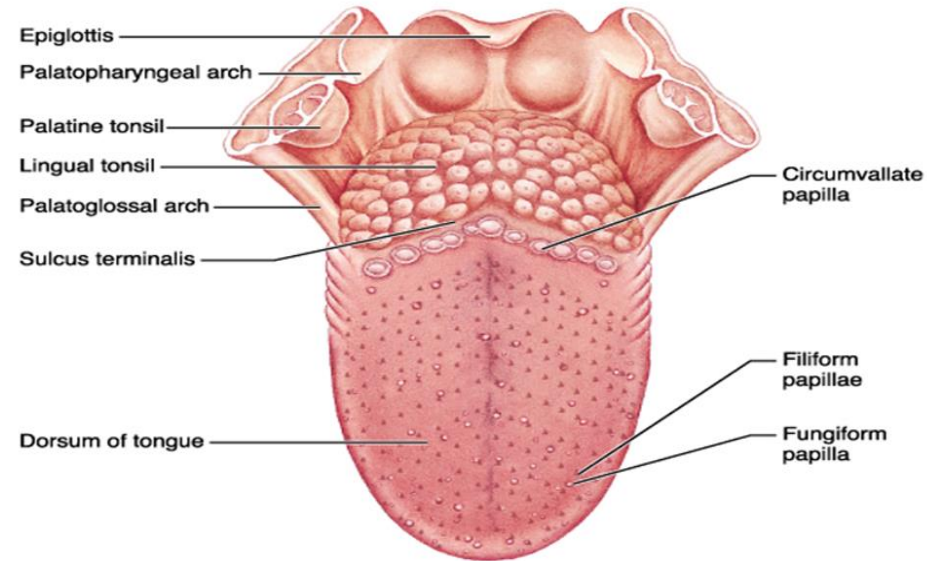
- **ventral surface** = the floor of the oral cavity covered by non-keratinized s.s.ep.
- **dorsal surface** is partly keratinized numerous small projections (papillae) cover the superior (dorsal) surface. Each papillae consists of a lining epithelium & a core of connective tissue.
- The anterior two thirds of the dorsal surface is separated from the posterior one third by a V-shaped groove
- Behind this there is lingual tonsils.



dorsal surface bears four types of papillae

1. Filiform –
2. Fungiform –
3. Circumvallate – V-shaped row in back of tongue
4. Foliate

- **Sulcus terminalis** – groove that separates the tongue into two areas:
 - Anterior 2/3 residing in the oral cavity
 - Posterior third residing in the oropharynx



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Filiform papillae

Are the smallest most numerous elongated conical shape , they present over the entire surface , the epithelium is partly keratinized give the tongue roughness. and provide friction aid in the processing of foods. Not contain taste buds.

Fungiform papillae

Mashroom shape ,occur singly and are located evenly between the filiform Their connective tissue core is richly vascularised. The epithelium is slightly thinner than on the remaining surface of the tongue. Contain scattered taste buds . Detecting sweet and salty taste.

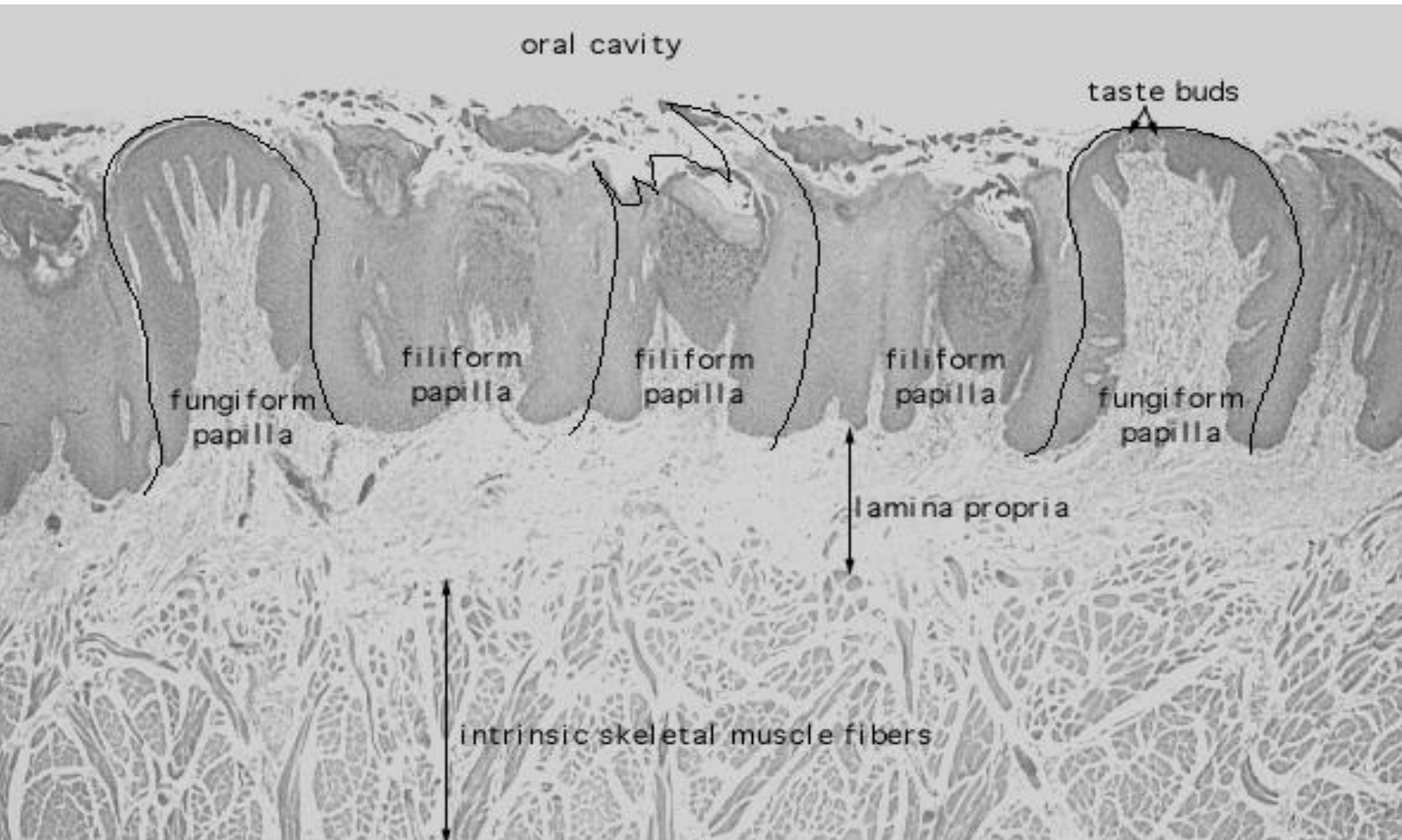
Filiform



Tongue

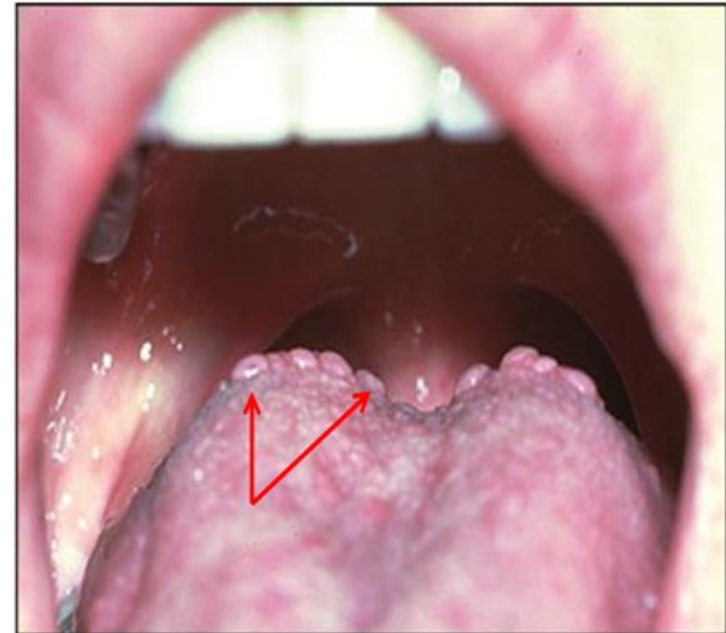


Tongue

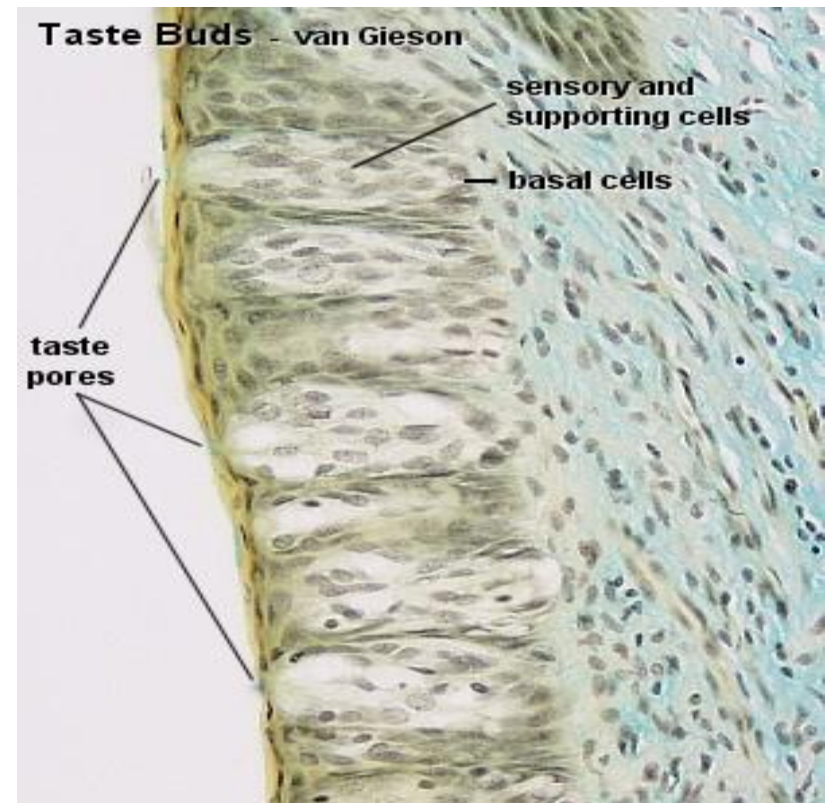
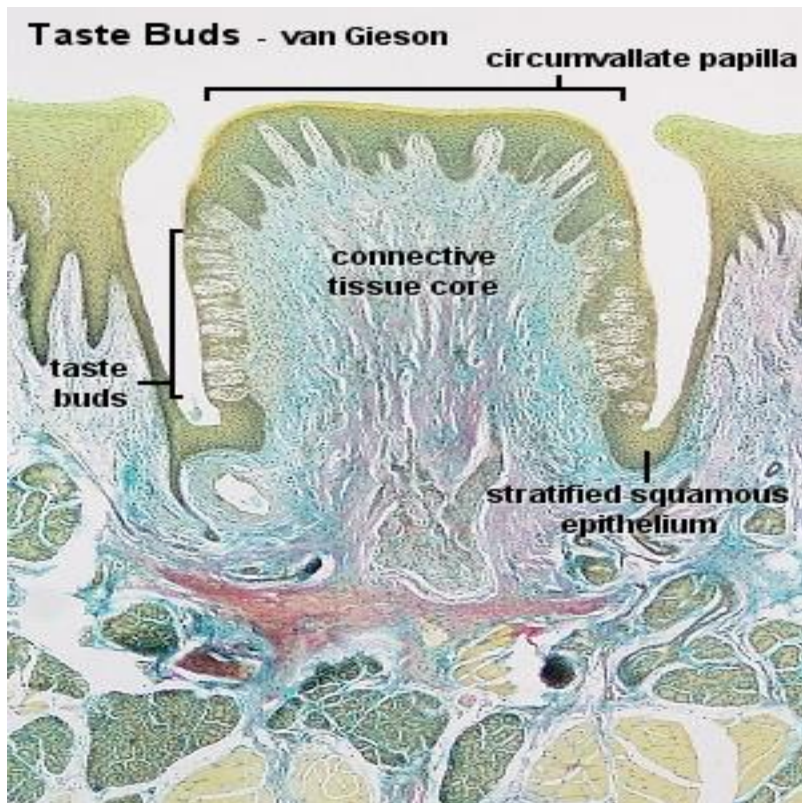


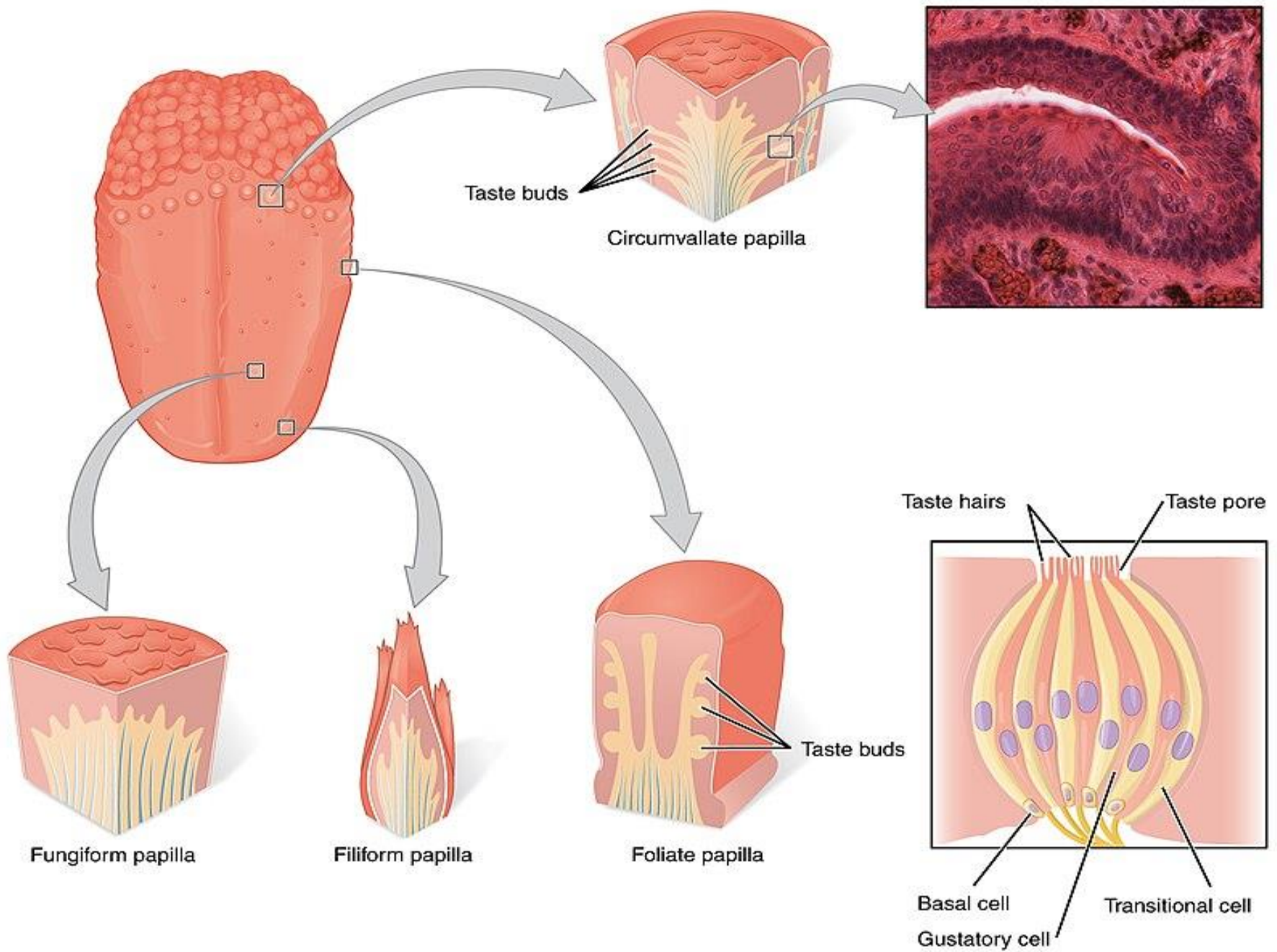
Circumvallate papillae

are largest in size and less numerous papillae - in humans there are between 8 and 12 of them. They are distributed in (V-shaped row in back of tongue), and are surrounded with a groove formed by the folding of the epithelium. Taste buds are particularly numerous on the lateral surfaces of these papillae. The excretory ducts of serous glands open into the grooves surrounding the papillae (glands of von Ebner). Detect bitter taste



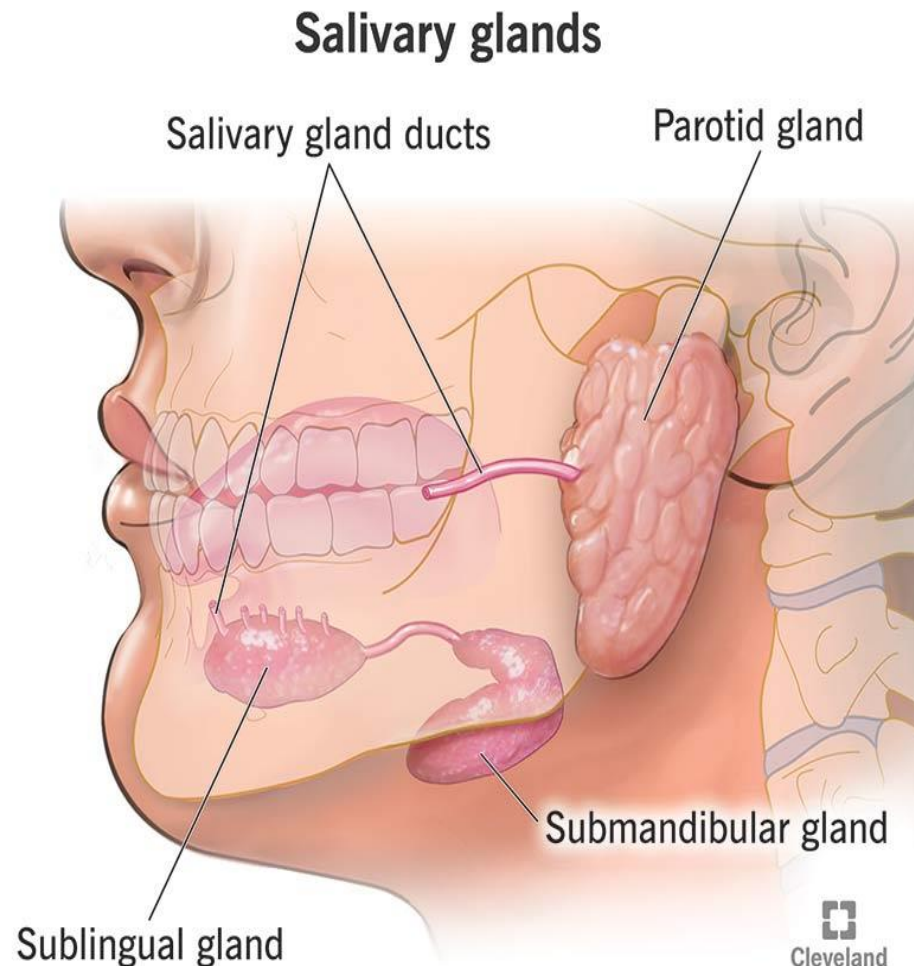
Taste Buds: Taste is a sensation performed by taste buds receptors located on the tongue
found in circumvallate papillae, fungi form papillae & foliate papillae.
In the smaller numbers on the soft palate, the epiglottis, palatoglossal arches & the posterior wall of the oropharynx.
sweet taste is best appreciated at the tip of the tongue, salt by the area just behind the tip & along the lateral border & bitter taste by circumvallate papillae.





Types of Salivary glands

- Collectively produce and secrete saliva.
- There are 2 groups of s.g.:
- **The major include:**
- 1-parotid
- 2-submandibular
- 3-sublingual
- **The minor :** numerous small glands located in the mouth

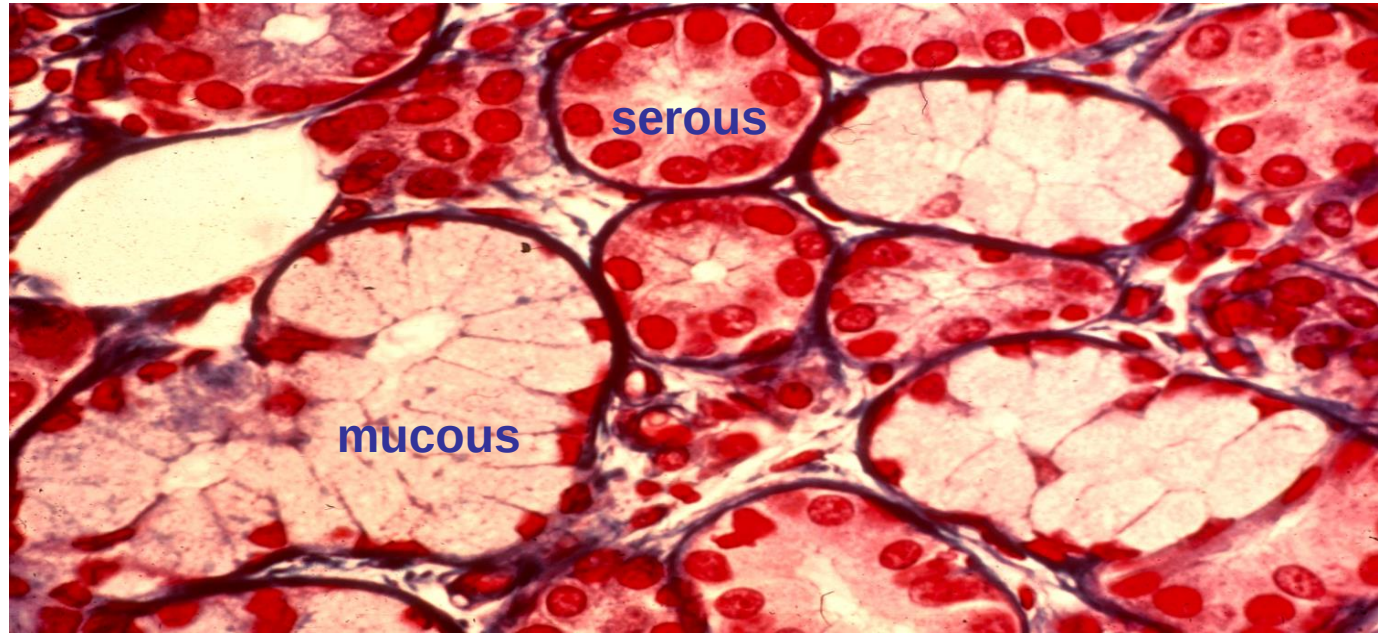


Major salivary glands

- All of them have capsule-connective tissue send septa divided into lobes—lobules carry blood vessels & nerves like branch of grapes
- Consist of :
- A-two general types of secretory cells
B- duct system

Cells lining the alveoli

- A-Secretory cells :1-Serous cells 2-Mucous cells
 - B-Myoepithelial cells
 - **Serous cells:**Are pyramidal in shape, usually form a spherical mass of cells called acinus or alveolus with narrow lumen
 - Basophilic cytoplasm contain protein secretory granules
 - they produce protein watery secretion
 - **Mucous cells**Are usually cuboidal to columnar , arranged as tubules
 - are larger more acidophilic than serous
- secrete mucous



System of ducts

1-Intralobular:small
ducts found in
lobule they are
surrounded by
thin layer of
c.t.are 2 types:

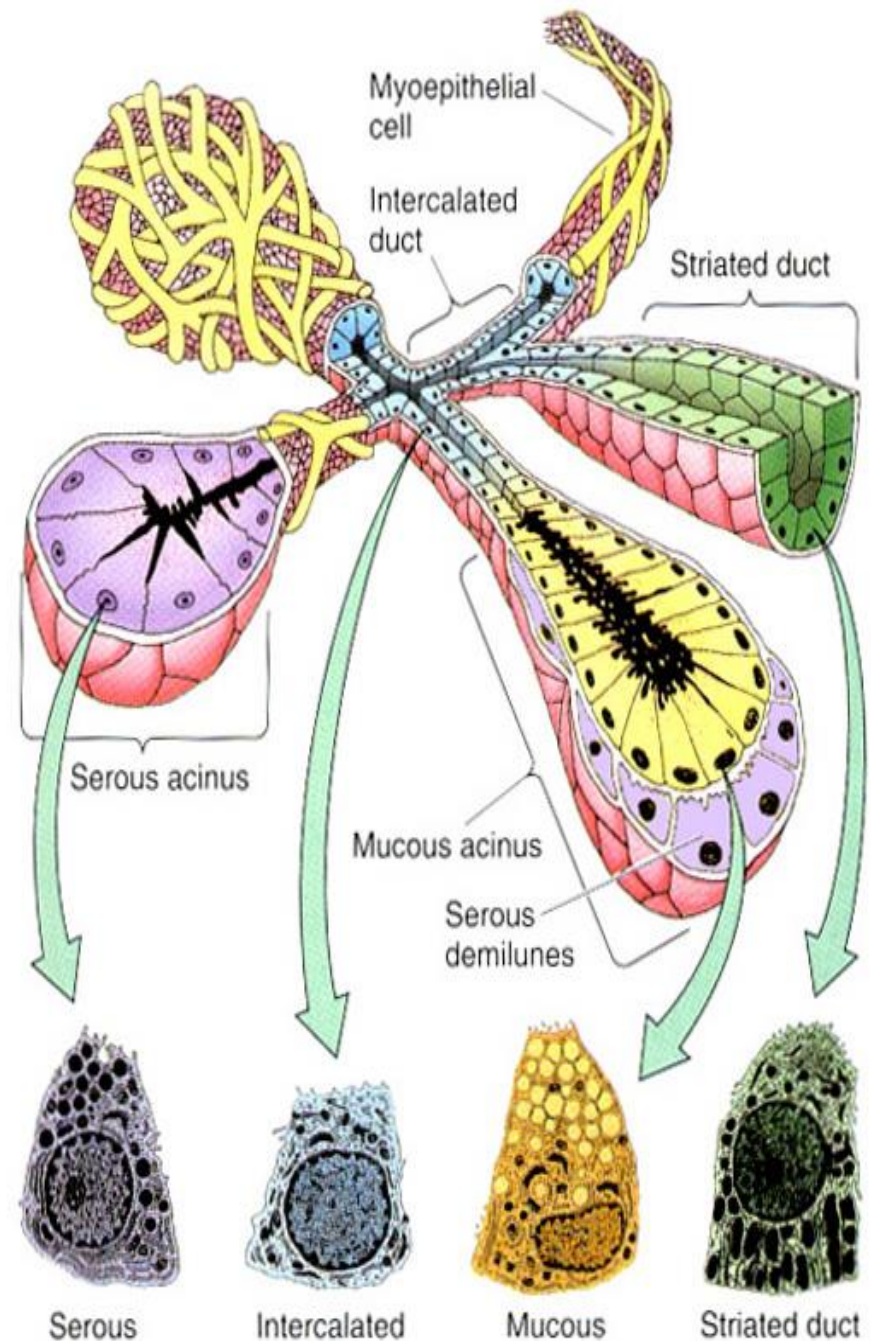
a-Intercalated ducts
narrow lumen

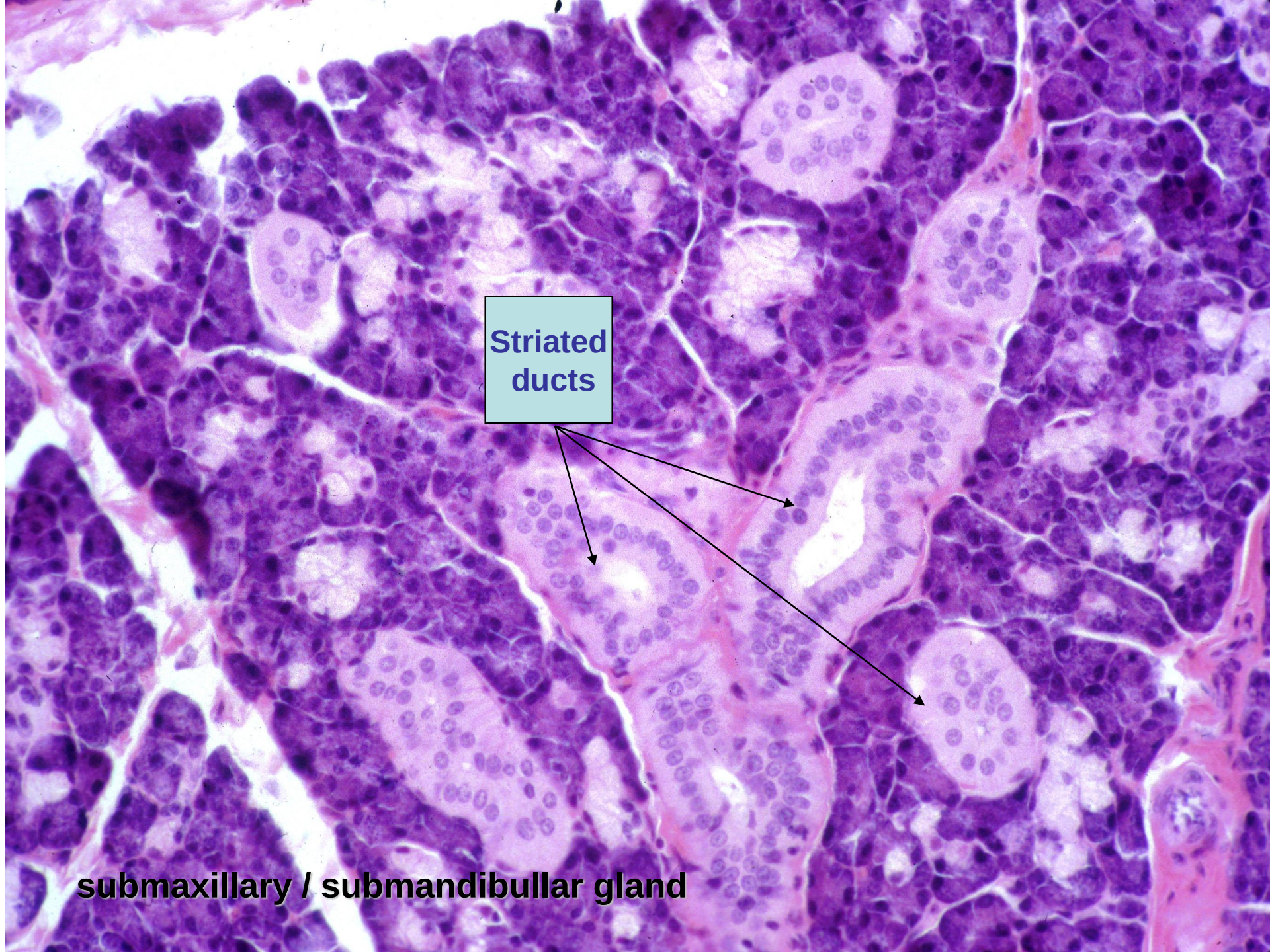
Simple cuboidal ep.

b-Striated ducts:

Has basal striation

Simple columnar



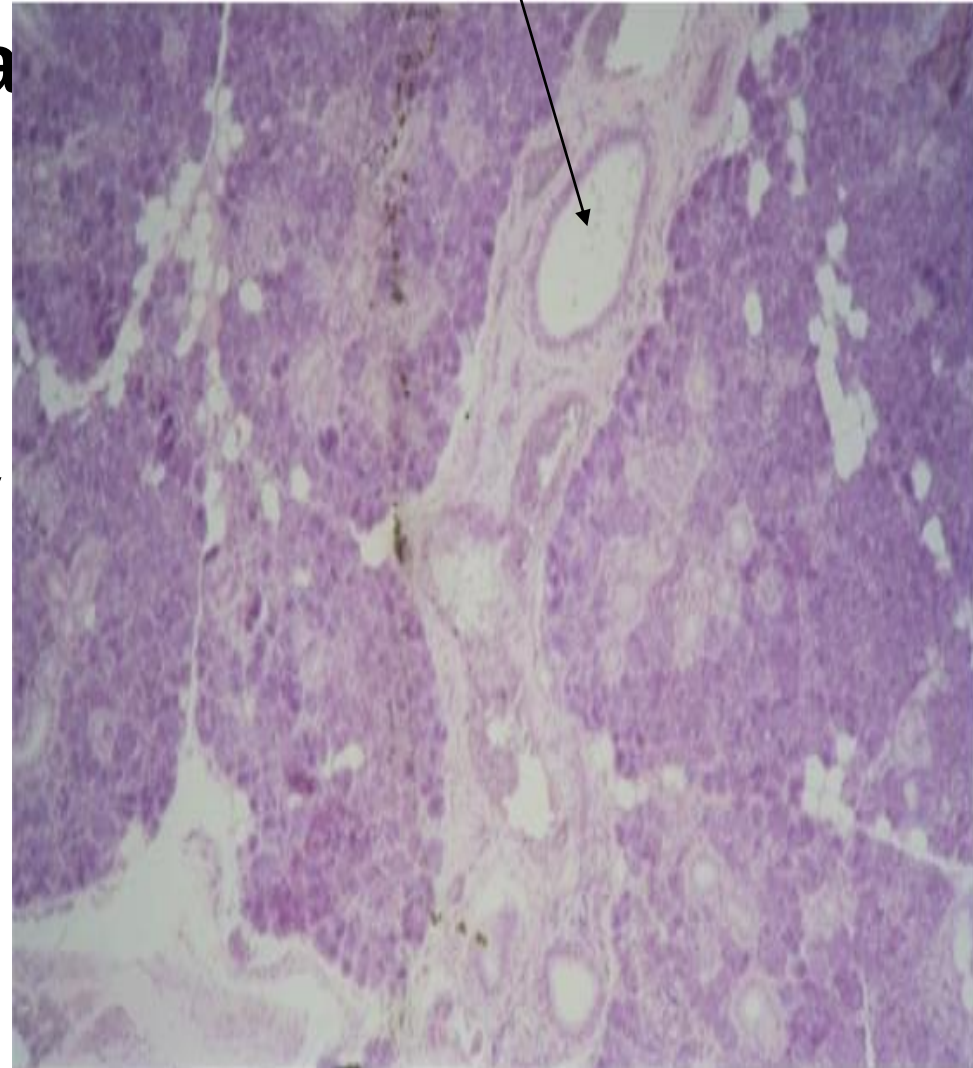


**Striated
ducts**

submaxillary / submandibular gland

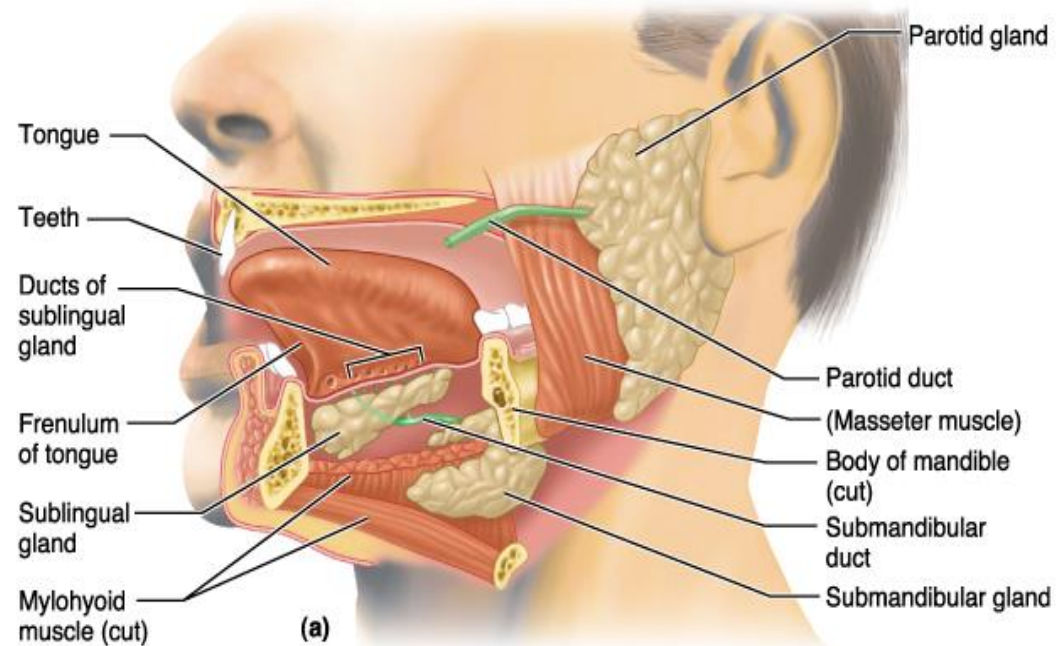
2-Interlobular ducts

- Located in the septa between lobules & lobes
- Larger than the intra. surrounded by large amount of c.t.
- The epithelia vary between simple columnar to stratified

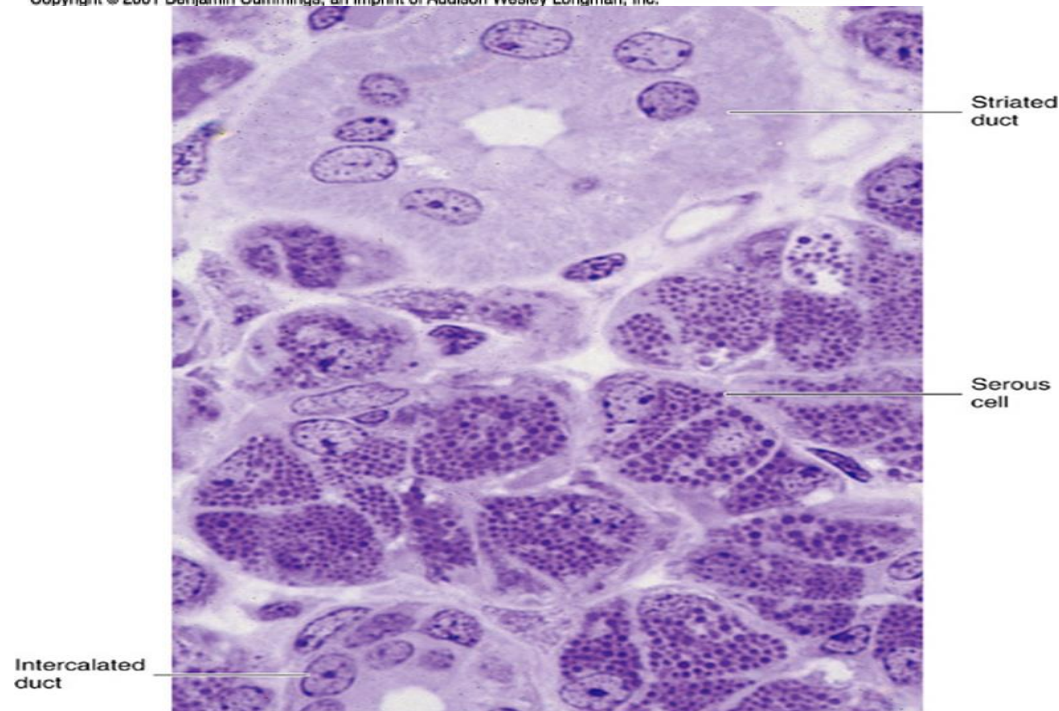


Parotid gland

- Largest salivary glands.
 - Beside the ear lies anterior to the ear
 - Their long one duct open into oral cavity opposite the second upper molar tooth on each side
 - Are compound acinar gland, are consist of serous
 - Produce about 25–30% of the saliva
- Long intercalated-
& short striated ducts.
-

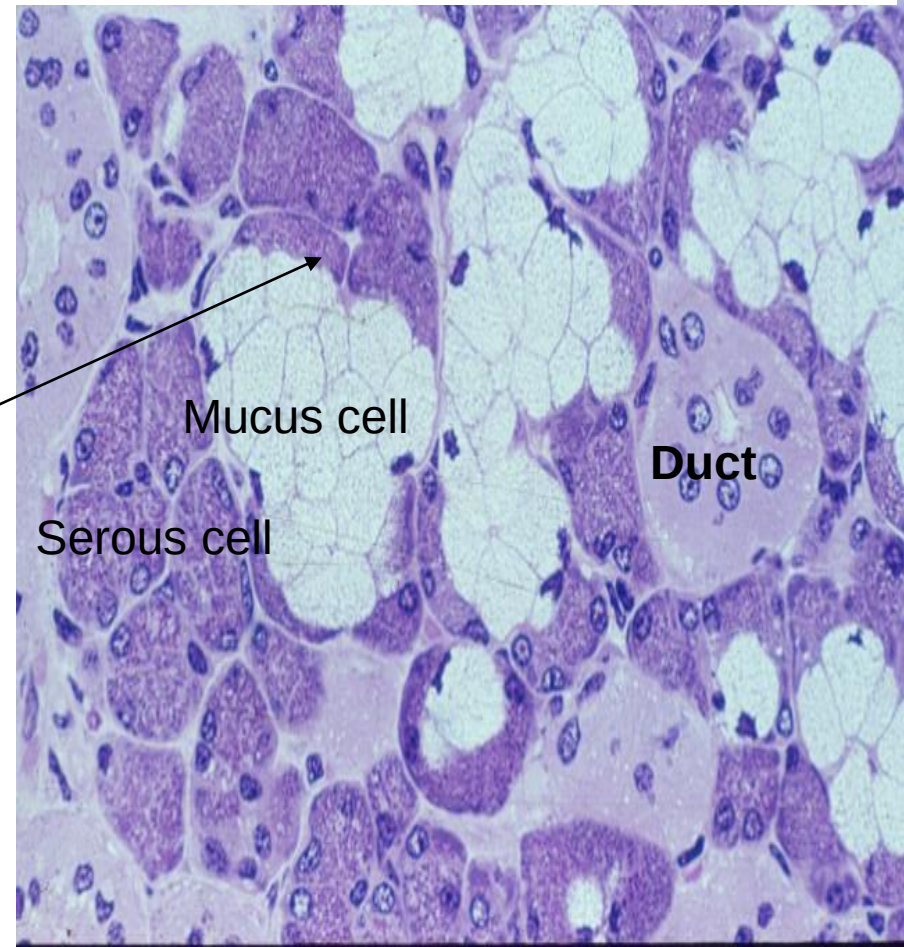
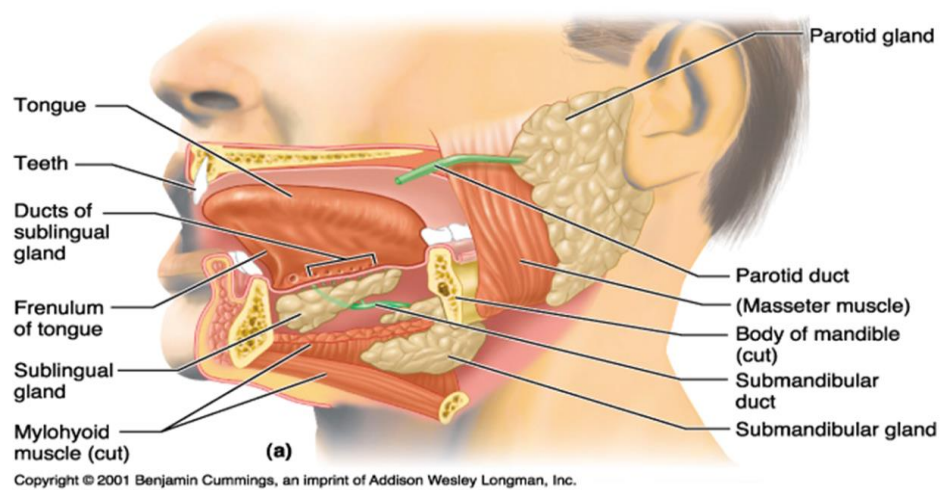


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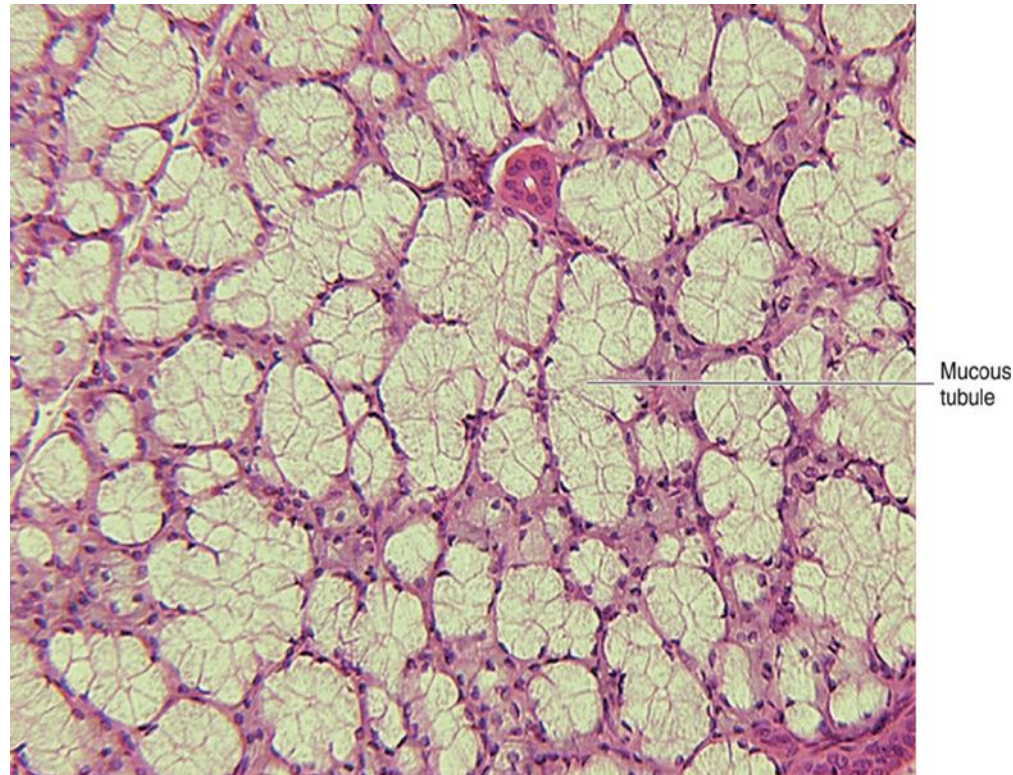
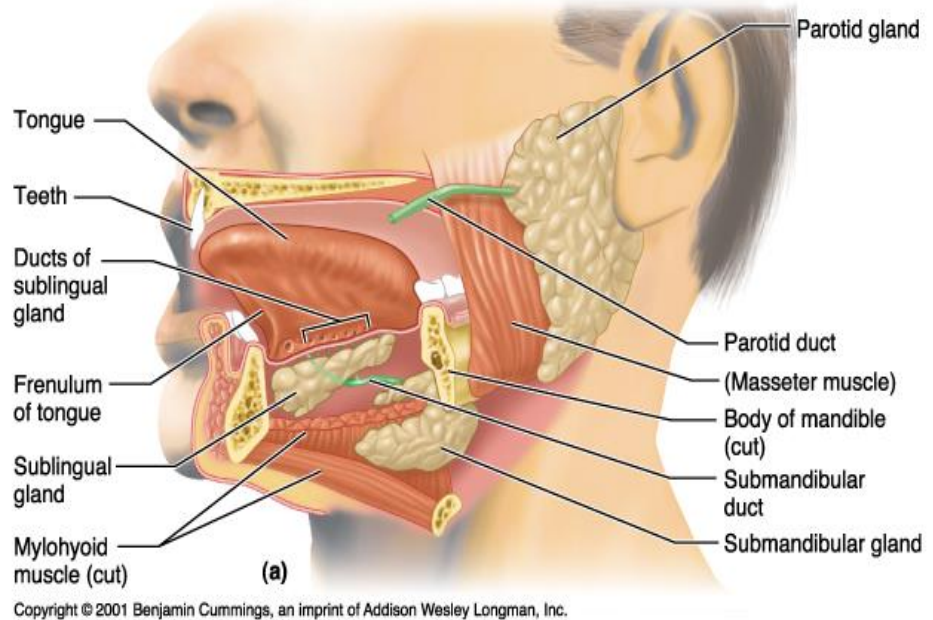
Submandibular gland :

- Compound tubulo-acinar
- Are ovoid shape
- Situated on the either side of neck just below the mandible
- Their duct open into floor of mouth one on each side of the frenulum of tongue
- Secretory portion contain both Serous-mucous, the serous is predominant
- 70%
- **Serous demilunes** secrete lysozyme
- Long striated while short intercalated ducts



Sublingual gland

- compound tubulo-acinar, are smallest glands
- Located in floor of mouth one on either side of the frenulum of the tongue
- each gland open into oral cavity through 10-12 small ducts
- Contain mucous & serous demilunes
- 5%
- Short intercalated, is ill defined & striated ducts



Thanks